

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405842
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Jentzsch; Eyck et al.

Apparatus and method for synchronizing participants of a simulation

Abstract

An apparatus for synchronizing participants of a simulation, including: a plurality of partitions, a backplane, and a synchronizer. Each partition is configured to run a respective simulation participant independently from other simulation participants. The backplane connects the plurality of partitions and is configured to transfer data between the plurality of partitions. The data include time stamps indicating a partition time of a simulation participant on at least one partition. The synchronizer is configured to synchronize the multiple partitions by stopping the running simulation participant of the at least one partition, when a time stamp of the running simulation participant is outside a synchronization window. The synchronization window has a predetermined time length indicating a range of acceptable time stamps (T1, T2) and starts with time (T0) associated with a time stamp of a slowest running simulation participant.

Inventors:	Jentzsch; Eyck (Neubiberg, DE), Koch; Andreas (Darmstadt, DE)
Applicant:	MINRES Technologies GmbH (Neubiberg, DE)
Family ID:	83902942
Assignee:	MINRES Technologies GmbH (Neubiberg, DE)
Appl. No.:	18/382086
Filed:	October 20, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240134718 A1	Apr. 25, 2024
US 20240231956 A9	Jul. 11, 2024

Foreign Application Priority Data

EP	22202781	Oct. 20, 2022
----	----------	---------------

Publication Classification

Int. Cl.: G06F9/52 (20060101)

U.S. Cl.:

CPC G06F9/52 (20130101);

Field of Classification Search

CPC: G06F (9/52); G06F (30/331)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6145099	12/1999	Shindou	714/E11.215	G06F 11/3636
9529662	12/2015	Wangkhem	N/A	G06F 11/0709
11321507	12/2021	Chang	N/A	G06F 30/3323
11327790	12/2021	Gaur et al.	N/A	N/A
2002/0162059	12/2001	McNeely	714/724	H04L 43/50
2002/0169863	12/2001	Beckwith	709/223	G06F 30/20
2008/0091403	12/2007	Harrison	703/17	G06F 30/33
2013/0124607	12/2012	Griffith	709/203	H04L 43/50
2013/0151220	12/2012	Matsumura	703/7	G06F 30/20
2015/0019193	12/2014	Boutillier	703/14	G06F 30/367
2017/0206247	12/2016	Gusev	N/A	G06F 16/2455
2019/0308104	12/2018	Nicolades	N/A	A63F 13/35
2020/0057707	12/2019	Jiang	N/A	G06F 11/261
2020/0272533	12/2019	Privitt	N/A	G06F 11/1637
2021/0365328	12/2020	Zimring	N/A	F16L 55/48
2021/0376994	12/2020	Zan	N/A	G06F 1/12
2022/0111517	12/2021	Bennice	N/A	B25J 9/1605
2022/0244978	12/2021	Dale	N/A	G06F 9/45558
2022/0374260	12/2021	Zhang	N/A	G06F 9/4843
2023/0076849	12/2022	Matsuoka	N/A	G06Q 10/06316

OTHER PUBLICATIONS

Seok, Moon GI et al., “An HLA-based Formal Co-simulation Approach for Rapid Prototyping of Heterogeneous Mixed-signal SoCs”, IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences, (online), vol. E100 No. 7, Jan. 1, 2017, 11 pages. cited by applicant

Steinman, Jeff S., “Interactive SPEEDES”, Simulation Symposium, Apr. 1, 1991, Proceedings of the 24th Annual New Orleans, LA, USA, Apr. 1-5, 1991, Los Alamitos, CA, USA, IEEE Comput. Soc, pp. 149-158. cited by applicant

Wilsey, Philip A., “Time Warp Simulation on Multi-Core Platforms”, 2019 Winter Simulation Conference (WCS), IEEE, Dec. 8, 2019, pp. 1454-1468. cited by applicant

Background/Summary

PRIORITY CLAIM

(1) The present application claims priority to European Patent Application No. 22 202 781.5, filed on Oct. 20, 2022, which said application is incorporated by reference in its entirety herein.

FIELD OF THE INVENTION

(2) The present invention relates to an apparatus and a method for synchronizing participants of a simulation and, in particular, to a loosely coupled parallel simulation of a designs of a register transfer logic on reconfigurable logic devices using a time window synchronization.

BACKGROUND OF THE INVENTION

(3) When designing computer chips, the design shall be checked for functionality before a possibly very cost-intensive chip fabrication. Since software-based simulations are typically very slow, especially for more complex designs, an alternative would be to map the description of the hardware design at a register transfer logic (RTL) level onto reconfigurable logic devices such as field-programmable gate arrays (FPGAs), on which the designs can now run in a faster emulated fashion. However, even large currently available FPGAs do not have sufficient capacity to accommodate very complex chip designs in only one FPGA. Therefore, the design is divided among several FPGAs and the communication links are switched among them. However, such systems are very complex and therefore expensive.

(4) Therefore, there is a need to increase the productivity of the development of complex hardware/software systems and, in particular, a need to simulate designs while allowing an efficient communication between the components and to provide flexible possibilities to adapt the simulation to very complex systems.

SUMMARY OF THE INVENTION

(5) At least some of the above-mentioned problems are solved by an apparatus for synchronizing participants of a simulation according to the claims and a corresponding method according to the claims.

(6) The present invention relates to an apparatus for synchronizing participants of a simulation. The apparatus comprises a plurality of partitions, a backplane, and a synchronizer. Each partition is configured to run a respective simulation participant independently from other simulation participants. The backplane is configured to connect the plurality of partitions and is configured to transfer data between the plurality of partitions. The data may include time stamps indicating a partition time of a simulation participant on at least one partition. The synchronizer is configured to synchronize the multiple partitions by stopping the running simulation participant of the at least one partition, when a time stamp of the running simulation participant is outside a synchronization window. The synchronization window has a predetermined time length indicating a range of acceptable time stamps and starts with a time associated with a time stamp of a slowest running simulation participant.

(7) The synchronizer may be any synchronization device, which may also be implemented by a particular installed software in a data processing device. Moreover, the subject of the simulation is a particular chip design, wherein the chip might be partitioned into various functional components (e.g., the partitions of the simulation). According to embodiments, the design of the chip is simulated (tested) before it is manufactured in order to optimize the various components to achieve a maximal performance and/or to use minimal resources of/for the manufactured chip. The

synchronizer controls the timing of the simulation in all partitions and is accommodated in the backplane. Therefore, according to embodiments, the synchronization takes not place in the partitions (i.e., decentralized). Rather, a centralized synchronization for all partitions is performed in the backplane by utilizing a window so that each partition can proceed with the simulation only upon granting a respective request sent to the synchronizer.

(8) According to embodiments, the simulation may test a hardware before manufacturing (e.g. as a monolithic chip in silicon). The tested hardware may be a complex chip (e.g. a system on chip, SoC) which may be divided in subcomponents (partitions). During the simulation, to test the exemplary chip of interest, these subcomponents are realized or implemented in hardware such as an FPGA or in a software emulation. In the following, each of these components may also be called a device under test (DUT), because each partition may be seen as a means for testing of hardware components having, e.g., a designated purpose.

(9) Therefore, optionally, the backplane comprises the synchronizer and the plurality of partitions includes at least one of the following sets: a set of field programmable gate arrays, FPGA, a set of virtual platform simulation partitions.

(10) The apparatus may be or include a computer or a data processing unit. However, the apparatus is not a dedicated simulation processing unit, but may represent a heterogeneous complex system comprised of independently operating units (partitions) being in communication with each other through a generic backplane. In particular, the synchronizer is outside or external to the partitions. In addition, each partition may include its own synchronizer and/or a scheduler to provide a timing for jobs to be carried out on the respective partition. However, these synchronizers/schedulers are different from the synchronizer of the apparatus which facilitates a synchronous structure between the separate partitions of the exemplary heterogeneous complex system.

(11) Optionally, the synchronizer may be configured to issue an approval of a request for updating a simulation time of a simulation participant (only) when the updated simulation time is inside the synchronization window.

(12) Optionally, the synchronizer is further configured to allow a stopped simulation participant to resume its operation upon receiving an external transaction. For example, when the simulation shall be stopped, all partitions may be started again to gracefully exit thereafter.

(13) Optionally, the apparatus further comprises one or more messaging interfaces to provide connections between the partitions and the backplane. The messaging interfaces may include an output connector and/or an input connector for mapping a generic message structure of the backplane to a design specific interfaces of the respective partitions. Therefore, the messaging interface(s) may map a generic message structure to design specific interfaces (e.g. of the partitions). The generic message structure advantageously allows the communication between hardware designs with arbitrary interfaces (e.g. of various FPGA partitions) and may enable a seamless interaction with the virtual components modeled in software (e.g. of the virtual platform simulation partitions).

(14) According to embodiments, the mapping on hardware interface between a generic data structure and a design specific data structure is performed over many abstraction levels (e.g. between pin levels of the register transfer logic, RTL) on the partitions and the generic message structure utilized on the backplane.

(15) Optionally, each partition may comprise a respective messaging interface, wherein each messaging interface is configured: to receive, at the output connector at a beginning of the simulation, a name of the input connectors being possible destinations of messages; and/or to register the output connector at the input connector.

(16) Optionally, each input connector comprises an asynchronous buffer to buffer data and provide them when the respective partition needs the buffered data. The buffer at the interface may include a first-in-first-out (FIFO) structure, which will ensure that no data (e.g. transactions, messages) are lost.

- (17) Optionally, each partition comprises a scheduler. The scheduler may be configured to schedule jobs of the respective simulation participant based on time stamps independently from other simulation participants. The synchronizer may be configured to allow the scheduler to freely schedule the jobs as long as the time stamps are inside the synchronization window. Internally, the partitions may handle their jobs (threads) freely.
- (18) Optionally, the synchronizer is configured to cause at least one of the following: to define or to determine a number of simulation participants; to register each simulation participant; to align all simulation participants at a beginning and/or the end of the simulation.
- (19) Therefore, the participants are known to each other (e.g. by name). Similarly, all connectors (input and output) may register each other.
- (20) Optionally, the synchronizer is adapted to set the predetermined time length of the synchronization window to at most 25 ms or at least 5 ms or about 10 ms, or use such windows. However, these are only examples. The synchronizer may be free to set any predetermined value for the time length of the synchronization window. The size of the synchronization window can be selected freely and may be adapted to optimize the simulation according to the needs (e.g. with respect to the time). The synchronization window may, in particular, be a synchronously running window, which proceeds or moves with the speed of the slowest participant (slowest partition).
- (21) Embodiments relate also to a computer-implemented method for synchronizing a simulation running on multiple partitions of a heterogeneous complex system. The simulation comprises simulation participants on the multiple partitions. The method comprises: running in parallel the simulation participants on the multiple partitions; transferring data between the partitions, the data including time stamps indicating a partition time of a simulation participant on at least one partition; defining a synchronization window of predetermined time length indicating a range of acceptable time stamps; synchronizing the multiple partitions by pausing the running simulation participant of the at least one partition, when a time stamp of the running simulation participant is outside the synchronization window.
- (22) According to further embodiment, one or more or all steps of the method may be carried out by a remote cloud computing unit. In addition, one or more steps may be repeated and the order of steps may be different to achieve the desired effect.
- (23) This method may also be implemented in software or as a computer program product. Embodiments of the present invention can, in particular, be a data processing system or may be implemented by software or a software module in a computing unit. Therefore, embodiments relate also to a computer program product or computer-readable storage device having a program code for performing the method, when the program code is executed on a processor.
- (24) Embodiments overcome at least some of the above-mentioned problems by a heterogeneous parallelization allowing a simulation of complex systems built from several subsystems. Advantageously, the subsystems are well encapsulated, have clear interfaces and relatively little communication between them. This makes it possible to partition a system at subsystem boundaries and flexibly emulate the individual partitions, for example, on multiple FPGAs. Embodiments allow to run a part of the design partly as hardware emulation (e.g. on reconfigurable logic devices) and partly as software simulation. Moreover, embodiments provide an infrastructure not only supporting off-the-shelf FPGA expansion cards for local on-premise simulation, but can also be deployed in the Cloud (on a remote cloud computing device). Likewise, embodiments provide an infrastructure for communication and synchronization of multiple FPGAs.
- (25) Particularly, instead of the proprietary prototyping or emulation systems of conventional systems, embodiments implement a generic synchronization logic that connects the different parts (partitions) of the partitioned chip design via a (generic) bus system ('backplane'). This allows to couple an arbitrary number of partitions implemented by exemplary FPGAs and/or virtual software emulations without providing fixed interconnections between the partitions. Both industrially available, relatively inexpensive FPGA accelerator cards and inexpensive cloud resources can be

used for this.

(26) Embodiments provide various advantages. These includes, inter alia, the scalability using FPGA resources that can be acquired at low cost or be utilized even in a rental model. Thus, the costs can significantly be reduced when compared to known prototyping/emulation systems of comparable power and performance. Moreover, the split between FPGAs and software emulation provides the opportunity to distribute the simulation participant in accordance to their performance. A slow running simulation participant may run on a FPGA partition and a fast running simulation participant may be implemented on a software emulation partition, or vice versa. The system provides all flexibility needed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various embodiments of the present invention will be described in the following by way of examples only, and with respect to the accompanying drawings, in which:
- (2) FIG. 1 depicts an apparatus for synchronizing participants of a simulation according to an embodiment system.
- (3) FIG. 2 depicts an infrastructure utilized by embodiments.
- (4) FIG. 3A, 3B depict further details of the messaging interface implemented in further embodiments.
- (5) FIG. 4 schematically shows a flow chart of method according to embodiments.
- (6) FIG. 5 illustrates the synchronization of FPGA partitions utilizing a synchronizer according to embodiments.
- (7) FIG. 6 illustrates a thread synchronization of virtual platform simulation partitions according to embodiments.
- (8) FIG. 7 illustrates the detailed synchronization flow of an exemplary SystemC partition according to embodiments.
- (9) FIG. 8 illustrates an integration of the FPGA partition and an execution flow according to embodiments.
- (10) FIG. 9 shows an architecture for an FPGA partition according to embodiments.

DETAILED DESCRIPTION

- (11) Various examples will now be described more fully with reference to the accompanying drawings in which some examples are illustrated.
- (12) Accordingly, while examples are capable of various modifications and alternative forms, the illustrative examples in the figures will herein be described in detail. It should be understood, however, that there is no intent to limit examples to the particular forms disclosed, but on the contrary, examples are to cover all modifications, equivalents, and alternatives falling within the scope of the disclosure. Like numbers refer to like or similar elements throughout the description of the figures.
- (13) It will be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.).
- (14) The terminology used herein is for the purpose of describing illustrative examples only and is not intended to be limiting. As used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes” and/or “including,” when used

herein, specify the presence of stated features, steps, operations, elements and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components and/or groups thereof.

(15) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which examples belong. It will be further understood that terms, e.g., those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(16) FIG. 1 depicts an apparatus according to an embodiment for synchronizing participants of a simulation (e.g. of a chip design) that is partitioned into parts. The apparatus comprises a plurality of partitions **100**, wherein each partition **100** is configured to run a respective simulation part (participant) independently from other simulation participants. The simulation parts may run in parallel or simultaneously.

(17) The apparatus further comprises a backplane **200** that connects the plurality of partitions **100**. The backplane **200** is configured to transfer data **150** such as messages or transactions between the plurality of partitions **100**. The data **150** may include time stamps indicating a partition time of a simulation participant on at least one partition **100**. The backplane **200** may implement a bus system to facilitate interactions between the partitions **100**. It forms a basis for other parts and can be implemented both in hardware, e.g. via PCIe, or in software as a C++ library. The backplane **200** maintains information about all partitions **100** and thus enables the interconnection between them.

(18) According to embodiments, the backplane is generic (i.e. not application specific) and can be realized in hardware or software.

(19) The apparatus further comprises a synchronizer **300** to synchronize the multiple partitions **100**. For this, a running simulation participant of at least one partition **100** can be stopped, when a time stamp **T5** of the running simulation participant is outside a synchronization window **310**. The synchronization window **310** has a predetermined time length indicating a range of acceptable time stamps and starting with time **T0** associated with a time stamp of a slowest running simulation participant. The time length can be adjusted to a particular simulation (e.g. in a range between 1 ms and 100 ms), but within a given simulation it will be predetermined (e.g. constant or modified in a defined manner). Therefore, the participants of the simulation are synchronized by utilizing this synchronization window. As long as the participants stay “inside” the synchronization window, the simulation participants can run freely, but they will be blocked by the synchronizer from proceeding too fast (e.g., when a time stamp is too far in the future). The pace of the simulation may be set by the slowest simulation participant.

(20) The synchronization **300** may be implemented as separate component as shown in FIG. 1. However, advantageously, synchronization device **300** can also be integrated or being part of the backplane **200** which provides the advantage the backplane **200** can block any routing of data from/to participant and thus stops effectively the respective participant.

(21) The parallelization is based on the ability to divide the system, e.g. based on the architectural features like CPU's or other components that are as independent as possible. The resulting partitions **100** may define each a device under test (DUT) that are simulated in parallel, e.g. by running on an FPGA and/or as software thread on virtual emulation.

(22) FIG. 2 depicts an infrastructure utilized by embodiments. The infrastructure connects all partitions **110**, **120**, **130**, **140** with each other and enables exchange of data **151**, **152**, **153**, **154** and the synchronization between the individual partitions **110**, **120**, **130**, **140**. As an example, FIG. 2 shows for the partitions **100** only two FPGAs **110**, **120** and only two virtual platform simulations **130**, **140**. However, in general, there can be any number of FPGAs **110**, **120** and/or any number of virtual platform simulation **130**, **140** (i.e. the indicated numbers **N** can be 1, 2, 3, . . .). Likewise, both numbers can be different implying that the number of FPGAs **110**, **120** can be different from

the number of virtual platform simulation **130, 140**. The infrastructure comprises three main parts: a backplane **200**—which provides the infrastructure for inter-partition communication; messaging interfaces **400**—which provide connectors utilized to connect individual components **100** of the multi-partition simulation to and from the backplane **200**; and the synchronizer **300**, which controls the simulation time in all partitions.

(23) According to embodiments, the message interfaces **400** implementing the communication between the partitions **100** are generic message interface connectors **400** that map generic messages to design specific interfaces of the partitions **100**. The generic message structure allows the communication between hardware designs with arbitrary interfaces (e.g. of different FPGA partitions **110, 120**) and may enable a seamless interaction with the virtual components modeled in software (e.g. of the virtual platform simulation partitions **130, 140**).

(24) In this embodiment, the synchronizer **300** and the backplane **200** are formed as one unit.

(25) FIG. 3A, 3B depict further details of the messaging interface **400**. All partitions **100** can be connected to the central backplane **200** via messaging interface connectors. For example, each messaging interface **400** may comprise an output connector **410** for sending data **150** and an input connector **420** for receiving data **150**.

(26) FIG. 3A illustrates the principles for an exemplary single connection which can be implemented directly using a single wire or a single bit. All connections between partitions **100** may be based on the same principles. According to embodiments, the inputs and outputs of the DUTs on the partitions **100** are connected to a specialized messaging interface connector **410, 420** and may include a module (sender) **412** that initiates a transaction (e.g., transmission of data **150**) by sending the data **150** through the output connector **410**. Data **151** from the output connector **410** are routed by the backplane **200** to the input connector interface **420** of the target module in another partition (not depicted in FIG. 3A).

(27) According to embodiments, the platform ensures an independence from specific hardware interface implementations. For this, a generic message structure (e.g. the data **151, 152**) is used for the communication via the backplane **200**. The specific hardware interfaces (e.g. on the partitions **100** receiving/forwarding the data **150**) are mapped to this generic structure. This mapping takes into account a possible need of crossing of several abstraction levels between pin levels of the register transfer logic (RTL) on the partitions **100** and the generic message structure utilized on the backplane **200**.

(28) In addition, more complex data structures may be transferred from a source partition (coupled to the output connector **410**) to a target partition (coupled to the input connector **420**). There, the transferred data is pushed into a target queue. For this, the input connector **420** may include an asynchronous FIFO buffer **422** (first-in-first-out queue) to handle the complex data structures with the generic message structure.

(29) The FIFO buffer **422** may be formed directly at the entry of the target partition to prevent data from getting lost in case the source partition (coupled to the output connector **410**) produces data faster than the target partition (coupled to the input connector **420**) can process. This FIFO buffer **422** may be a multi-producer single-consumer implementation with variable size. This safely handles backpressure and ensures that no input data can get lost. To map a complex interface connection (e.g.: AXI bus, advanced extensible interface). The input connector **420** may further comprise a listener **424** configured to retrieve and to deliver the data from the FIFO buffer **422**.

(30) The listener **424** may poll the asynchronous FIFO **422** for available elements and may send them as signal.

(31) FIG. 3B illustrates a transfer of more complex data structures to another partition utilizing the generic message structure of the backplane **200**. Here, a data path may be needed in both directions. For example, a read request is sent in the forward path during a read access, and the read data returns via the backward path. To achieve this, according to embodiments, the output connector **410** includes backward asynchronous FIFO buffer **416** and a transmitter **414**. The

transmitter **414** may pack data from an input socket, may send it to the backplane **200** and may poll the backward FIFO **410** for responses. The input connector **420** may be formed as in FIG. 3A. However, the listener **424** is also configured to forward the data on the backward path from input connector **420** to the backplane **200** which is then routed to the backward asynchronous FIFO buffer **416** of the input connector **410** where the data is available for delivery to the return address of the read request.

(32) For this, both sides of the connection should know about each other. According to embodiments, the connection is name-based. At the beginning of the simulation, an output connector **410** may receive the name of the destination of the message or data. This setup the forward direction. Before the output connector **410** starts a transaction, it registers itself at the destination connector (the corresponding input connector **420**). In this way, the backward path will be established. In particular, all connectors **410**, **420** can register each other so that each connector **410**, **420** knows the names and address of all other connectors of the other partitions.

(33) According to embodiments, the output/input connectors **410**, **420** (Signal and Socket) can be used to cover all common hardware interfaces like AXI (advanced extensible Interface), AHB (advanced high-performance bus), SPI (serial peripheral interface) etc. Multiple connections between a single pair of partitions can be allowed and can be implemented by an instantiation of multiple pairs message interface connectors.

(34) In order to ensure correct communication between several parallel-running components **100**, the partitions **100** need to be synchronized.

(35) FIG. 4 schematically shows schematically a flow chart of method **400** for synchronizing a simulation running on multiple partitions **100** of a heterogeneous complex system, wherein the simulation comprises simulation participants on the multiple partitions **100** that are connected by a backplane **200**. The method comprises: running **S100** in parallel the simulation participants on the multiple partitions **100**; transferring **S150** data **150** between the partitions **100**, wherein the data **150** includes time stamps **T1**, **T2**, . . . indicating a partition time of a simulation participant on at least one partition; defining **S230** a synchronization window **310** of predetermined time length indicating a range of acceptable time stamps **T1**, **T2**, . . . ; and synchronizing **S300** the multiple partitions **100** by pausing the running simulation participant of the at least one partition, when a time stamp **T5** of the running simulation participant is outside the synchronization window **310**.

(36) Therefore, embodiments utilize the synchronization window **310** concept for loosely timed coupling of partitions **100**.

(37) The individual partitions **100** running in parallel do not necessarily have to be FPGAs **110**, **120**. Parallel to the hardware FPGA partitions **110**, **120**, one or more partitions with hardware emulations **130**, **140**, for example a SystemC Virtual Prototype, can also be connected to the backplane **200**.

(38) A person skilled in the art knows SystemC as an extension of the programming language C++. It can be used for simulations, because it allows to create an executable model of a system of an as-yet undefined combination of hardware, software and data flows between them. SystemC is implemented as a class library to add hardware modelling constructs, such as concurrency, clocks, modules and interconnections, to the core C++ language.

(39) The usage of the synchronization window **310** allows to reduce the synchronization over-head. All schedulers within the partitions **100** can operate freely within the synchronization window **310**. However, they are never more ahead or behind each other than allowed by the synchronization window parameter (e.g. its predetermined length). For example, if the synchronization window **310** is, according to embodiments, set to 10 ms no two partitions will ever be separated by more than 10 ms in time.

(40) According to embodiments, all partitions **100** are connected to the synchronizer **300**. The synchronization of the different simulation participants **100**, both FPGA **110**, **120** and the exemplary SystemC **130**, **140**, may be organized in a way where the different partitions run relative

to each other in time. The progress of simulation time in individual partitions is controlled by the synchronizer **300**.

(41) According to embodiments, if a kernel on a partition **100** wants to update the current simulation time, it may get first an approval for the next time step by the synchronizer **300**. The synchronizer **300** keeps the current times of all partitions **100** within the synchronization window **310**.

(42) If a partition **100** requests a time step **T5** that is further in the future than the current simulation time **TO** of the slowest partition plus the value allowed by synchronization window **310** (cp. FIG. 1), the partition **100** will be put on hold and waits until the time stamp **T5** is within the synchronization window **310** or until an external transaction arrives and wakes-up the partition **100**. If any partition **100** stops the simulation, all sleeping partitions will be woken up and may gracefully finish the simulation.

(43) The synchronization window **310** is a user parameter and should be determined by design analysis or experimentally. The value of the synchronization window **310** may be defined at the beginning of the simulation. This mechanism provides the advantage of allowing the user to decouple the time between components in a controlled way. The synchronizer **300** implements a singleton that is atomically accessed by all kernels from all partitions **100**.

(44) Furthermore, according to embodiments, the number of partitions **100** is defined at the simulation start. This makes it possible to align all partitions at the beginning of the simulation. Otherwise it could happen that one partition is already finished before the other one is started.

(45) During the simulation, whenever a partition is going to update the current simulation time, embodiments may use the backplane interface **200** to get the next time step approved by the synchronizer **300**. The multi-partition simulation may run until one of the partitions **100** send a stop command to the synchronizer **300** (in the following also simply called synchronizer).

(46) According to embodiments, each partition may register automatically at the synchronizer **300**, may align with the other partitions **100** (so that all partitions begin the simulation at the same time) and may start the execution process.

(47) The following figures illustrate further specific details of the method for synchronizing.

(48) FIG. 5 illustrates synchronization of a FPGA partition **110** utilizing the synchronizer **300** according to embodiments. The dashed lines **111** indicates the enforced joined starting line (alignment) of the different partitions when the simulation starts. Similarly, the dashed line **112** indicates the end barrier where the simulation terminates. Between the synchronization points (dashed lines **111**, **112**) of the FPGA partition **110** and/or of the synchronization **300** may run freely.

(49) At step **S511**, the FPGA partition **110** starts data processing on the exemplary FPGA partition **110**. The first step in the execution loop synchronizes with other components (e.g. to achieve the desired alignment). For this, at step **S513**, the FPGA module/device **110** asks the synchronizer **300** for how long it is allowed to run and may request an approval of the next time step. At step **S530**, the synchronizer module **300** provides a time stamp, optionally together with the employed synchronization window **310**. For this, at step **S531**, the synchronizer **300** compares the requested time step request with the timing in other partitions **120**, **130**, **140** to make sure that all partitions remain in the set synchronization window **310**. At step **S532**, the synchronizer decides whether the next time step is bigger than the allowed time window (synchronization window **310**) or not. If so, at step **S533**, the next time step is adjusted to the window frame of the synchronization window **310** and, at step **S534**, the time step is saved and the new time stamp is returned to the requested partition **110**. If not, step **S533** is skipped and the system goes directly to step **S534**.

(50) Then, with the approval of the synchronizer **300**, the FPGA partition **110** sets the FPGA time (at step **S513**) and, at step **S514**, converts it into a number of cycles, which are executed at step(s) **S515**. After execution of the number of cycles, the running simulation of the FPGA partition **110** decides whether or not the simulation can continue (at step **S516**). The simulation on the FPGA partition **110** will continue until a stop signal is received from the synchronizer at step **S535**.

(51) If it shall stop, the FPGA partition **110** ends (finishes) the thread at step **S518**. If not, the simulation continues with reiterating steps **S513**, **S514**, **S515** and **S516** until the simulation shall stop.

(52) Therefore, according to embodiments, the FPGA module (partition **110**) asks the synchronizer **300** for how long it is allowed to run. The synchronizer **300** provides a time stamp and the DUT on the FPGA partition **110** may be executed for exactly the number of clock cycles to match the synchronizer time. The DUT execution period may then be blocked. As soon as it ends, the synchronization with other components (other partitions) will be performed again. This process is repeated until the synchronizer **300** stops the simulation. When there is nothing more to execute, the thread is put to sleep and is woken up either by a time update or by an input (e.g. a transaction package), or by the simulation stop.

(53) FIG. **6** illustrates a thread synchronization of a virtual platform simulation partition **130** (or partition **140**) which may be a so-called SystemC partition providing a software simulation of the corresponding DUT. As in FIG. **5** the simulation starts at step **S611**. At step **S612**, an initialization is performed, and a scheduler of a kernel of the virtual platform on the partition **130** is executed thereafter, at step **S613** (e.g. a delta cycle of the SystemC system). After this, at step **S614**, the scheduler decides whether the simulation time has finished or not. If so, the simulation exits at step **S617** which may be associated with a corresponding (external) stop signal received at step **S618**.

(54) If the simulation time at step **S614** has not yet finished, a time alignment is carried out at step **S615**, which again includes a request to the synchronizer **300** to get an approval for the new time stamp. If this is received, the simulation continues to advance in time at step **S616** and the delta cycle is again executed at step **S613**.

(55) The time alignment at step **S615** is performed similarly to the alignment as carried out for the FPGA partition **110** (see FIG. **5**). During this procedure, the synchronizer **300** compares, at step **S631**, the requested time step with time steps of the other partitions **110**, **120**, **140**. Thereafter, at step **S632** the synchronizer **300** decides whether the next time step is bigger than the allowed time window **310**. If so, at step **S633**, the simulation is set to a sleep until the time stamp is within the allowed synchronization window **310**. Thereafter, the new time stamp is saved and returned to the partition **130** at step **S634**. If the synchronizer **300** decides that the next time step is not bigger than the allowed time window **310**, step **S633** of placing the simulation to a sleep is bypassed and the simulation goes directly to the return of the new time stamp to the time alignment step **S615** of the partition **130**.

(56) This exemplary synchronization of a SystemC partition will ensure correct communication between several parallel-running components (partitions **110**, **120**, **130**, **140**). For example, an Accellera SystemC kernel may not provide this functionality (Accellera SystemC is a reference implementation of the SystemC language). If, according to embodiments, the SystemC kernel wants to update the current simulation time, it may have the next time step approved by the synchronizer **300**. The synchronizer **300** keeps the current time of all partitions **110**, **120**, **130**, **140** and a synchronization window **310**.

(57) For this, according to embodiments, the conventional Accellera SystemC scheduler may be extended to interact with the synchronizer **300**. The Synchronizer **300** intervenes in the scheduler loop always before the timestamp is updated at step **S616**. Then (see step **S631**) the “desired” timestamp is compared with the current timestamps in the other partitions **110**, **120**, **140** so that the synchronization window **310** is not violated. If the desired timestamp cannot be reached without violating the synchronization window **310**, the thread is put to sleep (at step **S633**) and woken up again as soon as the timestamp is allowed by the synchronizer **300**.

(58) FIG. **7** illustrates more details of the synchronization flow of the exemplary SystemC partition **130**.

(59) Again, the simulation starts at step **S711**, followed by step **S712** with an evaluation and an update for the simulation time. At step **S713**, the next time stamp is received. At step **S714**, the

system decides whether the simulation time is completed, and no external updates are available. If so, the simulation exits at step **S722**. If not, at step **S715**, the simulation decides whether the next time stamp is larger than the current time. If so, at step **S716**, the simulation again carries out a time alignment and, if not, the simulation proceeds with step **S719** (to process the timed event).

(60) This time alignment **S716** includes again, at step **S731**, a comparison of the time stamp request with the timing in other threads. At step **S732**, the synchronizer **300** decides if the next time step is bigger than the allowed time window **310**. If so, at step **S733**, the simulation is placed a sleep until the time stamp can be reached or until a transaction from another thread arrives. Thereafter, at step **S734**, the time step is saved and returned as a new time stamp to the thread. If the next time step is not bigger than the allowed time window at step **S732**, the step of placing the simulation at a sleep at step **S733** is bypassed and the new time stamp is immediately returned to the simulation.

(61) After the time alignment, at step **S717**, the time step is set and then, at step **S718**, the simulation decides whether any event is to be processed at the current time. If not, the simulation returns to step **S713**, and if so, the simulation processes, at step **S719**, the further events. Thereafter, at step **S720**, the simulation decides whether or not further runnable processes are available. If no runnable processes are available, the simulation completes at step **S721**. When there are still runnable processes, the simulation returns to step **S713** or to step **S718**.

(62) After completion of the simulation, at step **S721**, the simulation exits at step **S722** if the simulation time is completed. If not, the simulation proceeds with step **S712**.

(63) According to embodiments, when the exemplary SystemC kernel exits the simulation on the partition **130**, all other partitions **110**, **120**, **140** will also shut down properly. Again, according to embodiments, an Accellera SystemC scheduler is extended to interact with Synchronizer **300** as set out before.

(64) FIG. **8** illustrates an integration of the FPGA partition **110** and an execution flow. In the following, the term FIFO **422** includes a corresponding controller that manage the data buffering.

(65) First, at step **S811**, the FIFO **422** is checked for outstanding input transactions from other partitions **120**, **130**, **140** to the DUT on the exemplary FPGA partition **110**. If, at step **S812**, it is decided that there are any, they are transferred to the FPGA memory and the FPGA is adapted to locate the corresponding memory. Concretely, this is done by allocating buffer space on the FPGA partition **110** (at step **S816**) and to serialize and to write transactions to the buffer space (at step **S817**). The FPGA **110** is configured to have the address of the buffer space (step **S818**) available.

(66) Then the FPGA **110** is informed, at step **S813**, about the amount of cycles it may run. Internally (see steps **S831** to **S835**), the FPGA **110** will then update its expected simulation time (at step **S831**) and read the transactions from the memory (at step **S832**). While the FPGA runs until its simulation time matches the expected simulation time, drivers execute the transactions by applying the corresponding input signal transitions to the DUT (at step **S833**). Then, at step **S834**, monitoring devices watch the DUT's output signals to capture the transitions and convert them back into transactions. These transactions from the DUT are sent back to the Backplane (see next section). Monitored transactions are also written to a Ring Buffer (at step **S834**). When the current simulation time of the FPGA **110** reaches the expected simulation time it notifies the host about the completion of the task and returns the number of monitored transactions and their position in the Ring Buffer (at step **S835**).

(67) At step **S814**, the FPGA **110** verifies whether transactions are monitored. If so, at step **S819**, transactions are read from the buffer space (e.g. ring buffer, e.g. a direct memory access, DMA, buffer). At step **S820**, the transactions are converted from CBOR (concise binary object representation) to TLM (transaction level modeling). At step **821**, the transactions are sent to another partition **120**, **130**, **140** via FIFO. Therefore, the host reads these output transactions from the FPGA memory and sends them to the respective partition.

(68) At the end, at step **S815**, the method returns to signal the completion of the task to the backplane **200**.

(69) According to embodiments, the integration of the FPGA partitions **110, 120** is implemented as follows. First, there are two types of information which are to be exchanged between the infrastructure and the FPGA partitions **110, 120**: 1. Actual data transactions between partitions. 2. Timing information from the synchronizer **300**: up to which simulation time of the FPGA partition **110, 120** is allowed to run independently.

(70) Data transactions from other partitions **130, 140** to the DUT on the FPGA partition **110** are not directly sent, but instead are stored in a buffer **422** (e.g. a multiple producer single consumer, MPSC, FIFO as described with FIG. 3A, 3B). From there the data will be transferred to the FPGA partitions **110, 120** in batches. According to embodiments, this may be done each time the respective FPGA partition **110, 120** gets permission from the synchronizer **300** to continue the DUT execution (zoom-in into the “execute” block in FIG. 5).

(71) FIG. 9 shows an exemplary architecture for an FPGA partition **110**. The partition **110** includes a buffer **911**, two drivers **912, 913** (or more), a device under test **920** (hardware implemented by the FPGA), two monitors **930, 940** (or more), and an exemplary ring buffer **950**. The depicted hardware components may be utilized in the aforementioned methods.

(72) Transactions, for example, can either be generated by other partitions **120, 130, 140** and consumed by the DUT **920** on the FPGA **110** or generated by the DUT **920** and consumed by the other partitions **120, 130, 140**. Transactions generated by other partitions **120, 130, 140** will be handled by one of the drivers **912, 913** which converts them into signal transitions forwarded to the DUT **920**. For the transactions generated by the DUT **920** at least one of the monitors **930, 940** watches the outgoing signals and generates output transaction objects accordingly. For each interface of the DUT **920** which consumes transactions, a driver **912, 913** is created. For each interface which generates transactions, a monitor **930, 940** is created. Some interfaces (e.g. AXI) can consume as well as produce transactions. For these interfaces a driver **913** and a monitor **940** are created. Together they may form an agent **960**.

(73) Embodiments provide at least the following advantages. For example, the method and the system allow a parallel simulation of large systems divided among several FPGAs and/or software threads. This infrastructure enables a parallel simulation of RTL on FPGA and Virtual Platforms on a host. Likewise, it allows simulations of ASIC (ASIC=application specific integrated circuit) designs that are so large that they do not fit on a single FPGA. The parallel infrastructure can be run on-premise, but optionally also in a cloud (remote server).

(74) The description and drawings merely illustrate the principles of the disclosure. It will thus be appreciated that those skilled in the art will be able to devise various arrangements that, although not explicitly described or shown herein, embody the principles of the disclosure and are included within its scope.

(75) Furthermore, while each embodiment may stand on its own as a separate example, it is to be noted that in other embodiments the defined features can be combined differently, i.e. a particular feature described in one embodiment may also be realized in other embodiments. Such combinations are covered by the disclosure herein unless it is stated that a specific combination is not intended.

(76) Although the invention has been illustrated and described in detail by way of preferred embodiments, the invention is not limited by the examples disclosed, and other variations can be derived from these by the person skilled in the art without leaving the scope of the invention. It is therefore clear that there is a plurality of possible variations. It is also clear that embodiments stated by way of example are only really examples that are not to be seen as limiting the scope, application possibilities or configuration of the invention in any way. In fact, the preceding description and the description of the figures enable the person skilled in the art to implement the exemplary embodiments in concrete manner, wherein, with the knowledge of the disclosed inventive concept, the person skilled in the art is able to undertake various changes, for example, with regard to the functioning or arrangement of individual elements stated in an exemplary

embodiment without leaving the scope of the invention, which is defined by the claims and their legal equivalents, such as further explanations in the description.

LIST OF REFERENCE SIGNS

(77) **100** partition(s) **110, 120** reconfigurable logic devices (e.g. FPGAs) **130, 140, . . .** virtual platform simulation partition(s) **150** transferred data (e.g. messages) **200** backplane **300** synchronizer **310** synchronization window **400** messaging interface(s) **410** output connector(s) **412** sending device **414** transmitter **416, 422** (asynchronous) buffer (e.g. FIFO) **420** input connector(s) **424** listener **911, 950** buffer **912, 913** driver **920** device under test (DUT) **930, 940** monitors **960** agent **T0** time stamp of slowest running simulation part **T1, T2, . . .** time stamps inside the synchronization window **T5** exemplary time stamp outside the synchronization window **A1, A2, . . .** approval(s) of requests **R1, R2, . . .** update requests

Claims

1. An apparatus for synchronizing participants of a simulation of a chip design partitioned into parts, the apparatus comprising: a plurality of partitions, each partition being configured to run a respective simulation participant independently from other simulation participants; a backplane connecting the plurality of partitions and being configured to transfer data between the plurality of partitions, the data including time stamps indicating a partition time of a simulation participant on at least one partition; and a synchronizer configured to synchronize the multiple partitions by stopping the running simulation participant of the at least one partition, when a time stamp of the running simulation participant is outside a synchronization window, wherein the synchronization window has a predetermined time length indicating a range of acceptable time stamps and starts with a time associated with a time stamp of a slowest running simulation participant; wherein the synchronizer is configured to issue an approval of a request for updating a simulation time of a simulation participant, when the updated simulation time is inside the synchronization window; wherein all partitions are connected to the synchronizer which is configured to control the simulation time in all partitions; wherein each partition comprises a scheduler, the scheduler being configured to schedule jobs of the respective simulation participant based on time stamps independently from other simulation participants; and wherein the synchronizer is configured to allow the scheduler of the partitions to freely schedule the jobs as long as the time stamps are inside the synchronization window.
2. The apparatus according to claim 1, wherein the backplane comprises the synchronizer and the plurality of partitions includes at least one of the following sets: a set of field programmable gate arrays, FPGA, a set of virtual platform simulation partitions.
3. The apparatus according to claim 1, wherein the synchronizer is configured to allow a stopped simulation participant to resume its operation upon receiving an external transaction.
4. The apparatus according to claim 1, further comprising one or more messaging interfaces to provide connections between the partitions and the backplane, the messaging interfaces include an output connector and an input connector for mapping a generic message structure of the backplane to a design-specific interfaces of the respective partitions.
5. The apparatus according to claim 4, wherein each partition comprises a corresponding messaging interface, wherein each messaging interface is configured: to receive, at the output connector at a beginning of the simulation, a name of the input connectors being possible destinations of messages; and/or to register the output connector at the input connector.
6. The apparatus according to claim 4, wherein each input connector comprises an asynchronous buffer to buffer data and provide them when the respective partition needs the buffered data.
7. The apparatus according to claim 1, wherein the synchronizer is configured to cause at least one of the following: to define or to determine a number of simulation participants, to register each simulation participant, to align all simulation participants at a beginning of the simulation.

8. The apparatus according to claim 1, wherein the synchronizer is adapted to use as or to set the predetermined time length of the synchronization window to at most 25 ms or at least 5 ms or about 10 ms.

9. A computer-implemented method for synchronizing a simulation of a chip design partitioned into parts, the simulation running on multiple partitions of a heterogeneous complex system, the simulation comprising simulation participants on the multiple partitions, the method comprising: running in parallel the simulation participants on the multiple partitions; transferring data between the partitions, the data including time stamps indicating a partition time of a simulation participant on at least one partition; defining a synchronization window of predetermined time length indicating a range of acceptable time stamps; synchronizing the multiple partitions by pausing the running simulation participant of the at least one partition, when a time stamp of the running simulation participant is outside the synchronization window, wherein the synchronizer is configured to issue an approval of a request for updating a simulation time of a simulation participant, when the updated simulation time is inside the synchronization window, and wherein all partitions are connected to the synchronizer module which is configured to control the simulation time in all partitions; scheduling jobs of the respective simulation participant based on time stamps independently from other simulation participants by a scheduler comprised in each partition, and allowing the scheduler of the partitions to freely schedule the jobs as long as the time stamps are inside the synchronization window.

10. The method of claim 9, wherein at least one step is carried out by a remote cloud computing unit.

11. A non-transitory computer-readable storage medium with stored instructions adapted, when carried out on a data processing unit to carry out the method of claim 9.
